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COMPARISON AND ANALYSIS OF LEADING CLOUD SERVICE PROVIDERS (AWS, AZURE AND GCP)

Praveen Borra
Computer Science, Florida Atlantic University, Boca Raton, USA

ABSTRACT

Cloud computing services have helped many companies and organisations increase their efficiency and effectiveness in today's cutthroat business environment. Because of the proliferation of cloud computing service providers, this technology is evolving at a rapid pace. However, picking the right cloud provider in terms of the necessary service quality standards is still a huge hurdle. Popular cloud providers like as AWS, Azure, and GCP provide a variety of cloud services that are known for their expertise, dependability, and reasonable prices. Organizations across various sectors, including finance, communication, healthcare, business, and education, are increasingly drawn to these solutions. Three companies account for 70% of the cloud computing market: Microsoft Azure, Amazon Web Services (AWS), and Google Cloud Platform (GCP).

When selecting a cloud service provider to host websites or applications, many organizations and businesses choose between these three cloud service providers. Despite appearing similar at first glance, each provider has exceptional qualities and difficulties that significantly influence costs, ease of use, and overall experience. This paper outlines the major differences between AWS, Azure, and GCP, aiding organizations and businesses in making an informed decision about which cloud service provider to choose.

Keywords: Cloud Computing, Cloud leaders, AWS, AZURE, GCP, Cloud Service Providers

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I. INTRODUCTION

It is often impractical to invest substantially in hosting infrastructure, networks, servers, software, memory, and IT personnel to handle them, even though computerization is a need for all organisations and businesses in the present period. Computer resources can be automatically collected, used, and managed using cloud computing, which is based on an internet-based platform. Data centres that are part of cloud computing provide a wide variety of services to consumers and businesses by way of closely related resources that may be quickly and easily added or removed as needed. Depending on the cloud delivery model used, different sorts of services are offered. Software as a Service (SaaS), Infrastructure as a Service (IaaS), and Platform as a Service (PaaS) are the three main service models offered by cloud computing.

Infrastructure as a service (IaaS):

Infrastructure as a service (IaaS) allows users to access various parts of a cloud infrastructure, including storage, servers, networking gear, and virtualization layers, as needed. It gets rid of the hassle that businesses have with buying, managing, and maintaining technology on-premises. Instantaneous application deployment (IaaS) is best suited for experimental, short-term, or dynamic workloads.

It is often more efficient, faster, and cost-effective than on-premises deployments. Users deploy and run their own operating systems and software applications, with many cloud service providers offering this type of service.

Platform as a service (PaaS):

Software as a service (SaaS) provides a framework for running and managing applications by providing the underlying hardware, software, and development tools.

Like renting a vehicle, PaaS allows users to drive the development and operation of their software without the need to invest in or maintain the underlying infrastructure. Users create and run their software applications while relying on the cloud service provider for development tools, infrastructure, and the operating system. Examples of PaaS include Azure App Service, Azure Functions, Azure Logic Apps, AWS AKS, AWS Elastic Beanstalk, and Google BigQuery [25].

Software as a service (SaaS):

Users no longer need to install software locally thanks to SaaS, which offers cloud-based apps that are provided and managed by the cloud service provider. Software as a service allows users to access and use applications hosted in the cloud through the web. Amazon Chime, Office 365, and other email and calendaring services are examples of software as a service.

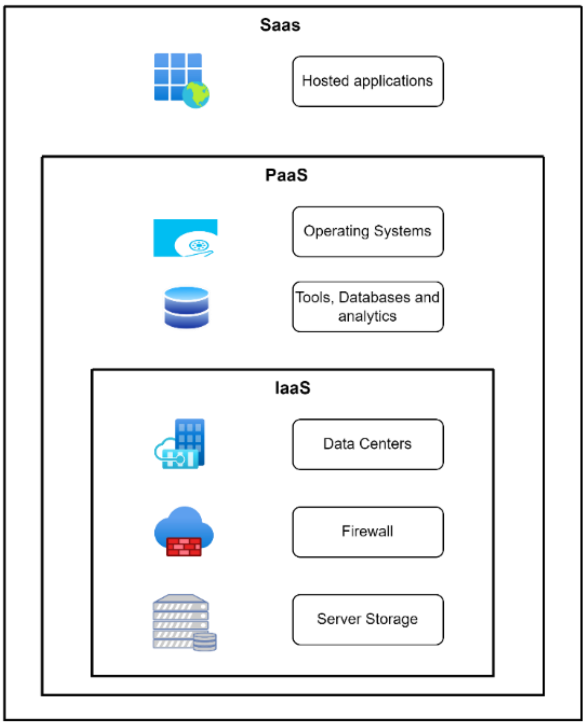

Figure 1: Difference between IaaS vs PaaS vs SaaS [25]

Public, private, and hybrid clouds are the three main varieties of cloud deployment models.

Public Cloud:

The computing resources and infrastructure used by public cloud architecture are owned and managed by third parties. The resources can be easily scaled up or down without investing much in new hardware or software because they are shared across several users.

Public clouds operate on multi-tenant architectures, serving multiple customers simultaneously.

Private Cloud:

In a private cloud architecture, a company owns and operates its own private cloud. Having it housed in-house in the company's data centre gives them more say over the resources and makes data and infrastructure safer.

Although private clouds provide increased control and security, they are generally more expensive than public clouds and require a higher level of IT expertise for maintenance.

Hybrid Cloud:

Hybrid cloud architecture allows for a versatile combination of public and private cloud services by combining aspects of both models. Based on their unique business needs and workload demands, organisations can easily move workloads between public and private environments using this strategy. Businesses that want some degree of data sovereignty while also taking advantage of public cloud services generally choose hybrid clouds.

Three prominent cloud computing systems, namely Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP), provide a vast array of services to both people and organisations.

Each platform has its strengths, features, and pricing models. AWS has traditionally held the largest market share, followed by Microsoft Azure, which has been gaining ground. Google Cloud Platform (GCP) is the third largest but has been steadily growing its presence in the market.

II. AMAZON WEB SERVICES (AWS)

Availability Zones and AWS Regions form the backbone of the AWS Cloud. Any given geographic area that houses more than one Availability Zone is known as an AWS Region. Distinct data centres with redundant power, networking, and connectivity are housed in distinct premises and make up each Availability Zone. With the help of these Availability Zones, companies and organisations may run databases and applications in production with better availability, fault tolerance, and scalability than they could with a single data centre.

Businesses and organisations can tailor their use of AWS's many cloud services to their specific requirements. Here we will go over the main AWS services in a categorised format. Analytics, Application Blockchain, Containers for Customers, Databases, Developer End-User Front-end, Business Cloud Compute, and Internet Machine Management are all services offered by AWS. News, Relocation, and Interim Technology in the realms of quantum robotics, storage, integration, applications, financial enablement, tools, web tech, identity management, mobile, the internet of things (IoT), delivery, compliance, and artificial intelligence (AI) [4].

The AWS Free Tier offers over a hundred different AWS products, each with one of three different types of free offers: free trials, free for 12 months, or always free.

AWS likewise gives both receptive and proactive technical support through different plans, including Developer, Business, Enterprise On-Ramp, and Enterprise plans. Organizations can select a support plan based on their specific business requirements.

How to access AWS services?

The AWS Management Console is an easy-to-navigate interface that users can use to access and administer Amazon Web Services (AWS). Users can also access resources on the move with the AWS Management Console Mobile App. One uniform tool for managing your AWS services is the AWS Command Line Interface (CLI). You can automate and manage various AWS services from the command line with a single utility that you just need to download and configure.

AWS CloudShell, accessible next to the search bar in the AWS Management Console, provides a browser-based shell that is pre-authenticated with your console credentials. Using CloudShell, you can quickly run AWS commands and scripts without leaving your web browser.

By providing an API that is specific to your language or platform, our Software Development Kits (SDKs) make it easy for your apps to integrate with AWS services.

III. MICROSOFT AZURE

Globally distributed Microsoft Azure resources are based on Availability Zones and Regions. In Azure, the world is divided into regions according to national borders or geopolitical lines. A geographical area that includes one or more data centres that are close by, connected, and have minimal latency is called a region. Azure data centres are physically separated into zones called availability zones. Data centres within an Availability Zone have their own cooling, power, and networking systems. With their built-in isolation features, all of the zones will keep running even if one of them goes down.

Private fibre optic networks that are very fast link the availability zones. You can utilise any combination of Azure's various cloud services to meet the specific demands of your company or organisation. Azure provides resources for artificial intelligence, machine learning, analytics, compute, containers, databases, devops, identity, integration, internet of things, media, migration, mobile, networking, security, storage, virtual desktop infrastructure, web, management and governance, and more [5].

To have access to Azure's services, organisations and businesses can sign up for a free Azure account. Users are given a $200 credit to be used within the first 30 days of signing up. Also, there are two types of services that customers may take advantage of at no cost: popular services, which are free for a year, and more than 55 other services, which are free forever. Microsoft Azure Cloud offers a range of support plans, including reactive and proactive technical help. Basic, Developer, Standard, and Professional Direct are the plans that are handed out. Companies can choose a support plan that works for them.

How to access Azure services?

The Azure portal provides a graphical user interface (GUI) for accessing and managing Microsoft Azure. For automated and command-line interactions with Azure, users can additionally utilise Azure PowerShell and the Azure Command-Line Interface (CLI). Users are able to automate many Azure services using scripts and control them from the command line with these tools.

Azure CloudShell, available next to the search bar in the Azure portal, provides a browser-based shell pre-authenticated with user console credentials. Azure CloudShell allows users to run Azure commands and scripts quickly without leaving their web browser.

Users can manage and keep tabs on all of their Azure accounts and resources from the convenience of their iOS or Android devices with the Microsoft Azure mobile app. Notifications and alerts on key health issues can keep organisations informed as they check the current status and important metrics of their services.

IV. GOOGLE CLOUD PLATFORM (GCP)

Virtual machines (VMs) and physical assets like hard drives and computers are both part of Google Cloud, which is located in data centres all around the world. You can choose among data centres in North America, South America, Europe, Africa, the Middle East, and Asia, among other places. Different zones within a territory are physically separated from one another. A combination of the region name and a letter identifier is used to identify each zone. Machine learning and artificial intelligence, business intelligence, compute, containers, data analytics, databases, developer tools, distributed cloud, industry specific, integration services, management tools, media services, migration, mixed reality, networking, operations, productivity, security, and identity are all part of Google Cloud Platform (GCP) [15]. New clients can take advantage of a $300 credit to test out Google Cloud services and create a proof of concept. This credit is valid for about 90 days.

Customers will only be charged once they activate their full paid account. The Free Trial includes Cloud Billing credits that cover the costs of resources used while familiarizing customers with Google Cloud [18].

Google Cloud Platform (GCP) provides both receptive and proactive technical support through various plans: Basic, Standard, Enhanced, and Premium. Organizations can choose a support plan that best aligns with their business requirements [19].

How to access GCP services?

Google Cloud offers three main methods to interact with its services and resources. You may manage your Google Cloud projects and resources through the web-based graphical user interface called the Google Cloud Console. The Google Cloud CLI is a command-line interface (CLI) that allows users to do most activities on Google Cloud. Additionally, Google Cloud offers client libraries that simplify the creation and management of resources.

V. COMPARISION BETWEEN AWS VS AZURE VS GCP

Cloud computing offers cost-effectiveness, scalability, high availability, robust security, greater flexibility, compliance, and enhanced monitoring compared to traditional on-premises infrastructure. AWS is the leading cloud platform globally in terms of usage, while Azure is intensifying the competition. GCP is experiencing rapid growth fueled by innovation [12].

	AWS	Azure	GCP
Launch Year	2006	2010	2008
Market Share	32	23	10
Availability Regions	30+	60+	40+
Countries	200+	200+	200+
Services	200+	200+	120+

Table 1: AWS vs Azure vs GCP [25]

Some of the key services offered by AWS, Azure, and GCP.

Compute Services:

There are a number of alternatives for hosting application code for organisations, including AWS, Azure, and GCP. In this context, "compute" means the hosting model of the application's resources.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Compute Instance	Amazon EC2 Instance	Azure Virtual Machines	Compute Engine
Applications, containers	AWS Lambda, AWS Fargate, AWS App Runner	Azure service app	App Engine
Functions, containers	AWS Lambda	Azure Functions	Cloud functions for Firebase
Containers	Amazon Elastic Kubernetes Service, Amazon Elastic Container Service	Azure Kubernetes Service	Google Kubernetes Engine
App modernization	AWS Bottlerocket	Azure Container Instances	Container-Optimized OS

Table 2: AWS vs Azure vs GCP Compute Services

Storage Services:

Virtual machine (VM) disc data storage is one of many services offered by AWS, Azure, and GCP.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Object storage	AWS Simple Storage Service (S3)	Azure Blob Storage	Cloud Storage
Infrequently accessed object storage	Amazon S3 Glacier	Azure Archive Storage	Cloud Storage Archive
File Storage	Amazon Elastic File System	Azure Files	Filestore

Table 3: AWS vs Azure vs GCP Storage Services

Networking:

You can use the several networking options offered by AWS, Azure, and GCP on their own or in tandem.

Services	Amazon Web Services (AWS)	Microsoft Azure Service	Google Cloud Platform (GCP)
Cloud virtual networking	Virtual Private Cloud (VPC)	Virtual Network (VNET)	Virtual Private Cloud
NAT gateway	NAT Gateways	Virtual Network NAT	Cloud NAT
DNS management	Amazon Route 53	Azure DNA	Cloud DNS
Dedicated network	AWS Direct Connect	Azure ExpressRoute	Cloud interconnect
Load balancing	Elastic Load Balancer	Load Balancer	Cloud load balancing
Content delivery network	Amazon CloudFront	Azure Front Door	Cloud CDN and Media CDN

Table 4: AWS vs Azure vs GCP Network Services

Security, identity, and access:

AWS, Azure, and GCP offer a variety of tools and capabilities to help create secure solutions on their cloud platforms. These services provide for clear accountability while guaranteeing the privacy, security, and accessibility of client data.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Security and identity	Identity and Access Management (IAM)	Microsoft Entra ID (Azure AD)	Cloud Identity
Security and identity	AWS Systems Manager	Azure Bastion Host	Identity-aware proxy (IAP) TCP forwarding
Security and identity	Amazon Cognito	Azure Active Directory B2C	Identity Platform

Table 5: AWS vs Azure vs GCP Security, identity, and Access Services

Messaging and eventing:

AWS, Azure, and GCP offer various messaging and eventing tools. Messaging enables applications and services to communicate through a central message broker. Eventing, also known as data streaming, involves continuously processing data within an infrastructure.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Event handling	AWS EventBridge	Azure Event Grid	Eventarc
Streaming data ingest	Amazon Kinesis	Azure Event Hubs	Pub/Sub
Messaging	Simple Notification Service (SNS)	Service Bus	Pub/Sub
Parallel task	Simple Queue Service (SQS)	Queue storage	Cloud task

Table 6: AWS vs Azure vs GCP Messaging and eventing services

Databases:

Cloud databases offer the optimal solutions for organizations building enterprise generative AI applications, regardless of size. AWS, Azure, and GCP give various database options to support these needs.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Relational Database	Amazon Relational Database Service (RDS)	SQL Database, Database for MySQL, Database for PostgreSQL, Database for MariaDB	Cloud SQL, Bare Metal Solution, AlloyDB for PostgreSQL
Serverless relational database	Amazon Aurora Serverless	Azure SQL Database serverless, Serverless SQL pool in Azure Synapse Analytics	Cloud Spanner
NoSQL	NoSQL, DynamoDB (Key-Value), SimpleDB, Amazon DocumentDB (Document), Amazon Neptune (Graph)	Azure Cosmos DB	Firestore, Datastore, Bigtable
Caching	ElastiCache, Amazon MemoryDB for Redis	Cache for Redis	Memorystore
Database migration	AWS Database Migration Service	Azure Database Migration Service	Database Migration Service

Table 7: AWS vs Azure vs GCP Databases services

Data warehouse:

A data warehouse is a centralized repository that stores processed operational data, metadata, summary data, and raw data for easy user access. AWS, Azure, and GCP provide various data warehouse options.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Data warehouse	Amazon Redshift	Azure synapse analytics	Big Query

Table 8: AWS vs Azure vs GCP Databases services

Big data processing:

To facilitate decision-making, big data processing makes use of a framework and set of methods that allow access to massive data sets and the extraction of useful information from them. AWS, Azure, and GCP provide various big data processing options.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Big data processing	Amazon Elastic MapReduce (EMR)	Azure Data Lake Storage (ADLS), HDInsight and Databricks	Dataproc

Table 9: AWS vs Azure vs GCP Big data processing services

Data orchestration / ETL:

Data pipeline orchestration involves the movement and integration of data from multiple sources to prepare it for analysis and delivery to end-users. ETL, which stands for Extract, Transform, and Load, refers to the process of constructing these pipelines. AWS, Azure, and GCP provide several data orchestration and ETL services.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Data integration / ETL	Amazon AppFlow, Amazon Data Pipeline, and AWS Glue	Azure data factory, Azure Synapse studio	Cloud Data Fusion
Data wrangling	AWS Glue Data Brew	Azure data factory, Azure Synapse studio	Dataprep by Trifecta
Workflow orchestration	Amazon Data Pipeline, AWS Glue and Managed Workflows for Apache Airflow	Azure data factory, Azure Synapse studio	Cloud Composer

Table 10: AWS vs Azure vs GCP Data integration/ETL services

Analytics and visualization:

Analytics and visualization are essential processes that aid users in comprehending and interpreting data. Data analytics employs systematic methods to uncover patterns, trends, and relationships within data. Data visualization transforms data into visual formats like charts, graphs, or maps, enhancing analysis and interpretation. AWS, Azure, and GCP provide a variety of analytics and visualization services.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Business intelligence	Amazon QuickSight	Microsoft Power BI	Looker
Stream data processing	Amazon Kinesis Analytics	Azure Stream Analytics	Data Flow
Data analytics	Amazon Athena	Azure Synapse Analytics	BigQuery

Table 10: AWS vs Azure vs GCP analytics and visualization services

Internet of Things (IoT):

Things that are part of the Internet of Things (IoT) are those that have sensors, software, and other technologies built into them that allow them to communicate with one another and other systems and devices over networks like the Internet. AWS, Azure, and GCP provide various data orchestration and ETL services.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
Internet of Things (IoT)	AWS IoT Greengrass	Azure IoT Edge	Edge TPU
Internet of Things (IoT)	AWS IoT Core	Azure IoT Hub	Cloud IoT Core

Table 12: AWS vs Azure vs GCP Internet of Things (IoT) services

AI and machine learning:

Machine learning (ML) and artificial intelligence (AI) are sometimes used interchangeably, but they really have distinct uses, datasets, and approaches. AWS, Azure, and GCP provide numerous AI and machine learning services.

Services	Amazon Web Services (AWS)	Microsoft Azure	Google Cloud Platform (GCP)
ML Platform	Amazon SageMaker	Azure Machine Learning	Vertex AI custom training
Bot	Alexa Skills Kit	Microsoft Bot Framework	Dialogflow
Speech recognition and synthesis	Polly, Transcribe	Speech services	Speech-to-Text, Text-to-Speech
Video AI	Amazon Rekognition Video	Azure Video Analyzer for Media	Video AI
Natural language processing	Amazon Comprehend	Azure Text Analytics	Natural Language AI
Translation	Amazon Translate	Azure Translator	Translation AI

Table 13: AWS vs Azure vs GCP AI and machine learning services

VI. BENEFITS OF AMAZON WEB SERVICES (AWS), MICROSOFT AZURE, AND GOOGLE CLOUD PLATFORM (GCP)

Amazon Web Services (AWS)

For more than a decade, AWS has served as the bedrock of Amazon.com's massive web-based enterprise, providing a worldwide computing infrastructure that is scalable, dependable, and secure. Auto Scaling and Elastic Load Balancing are two AWS features that allow applications to gradually scale up or down depending on demand. This makes use of Amazon's vast infrastructure for computing and storage resources. Members of the AWS Specialisation Partner programme have exclusive access to AWS roadmaps, get to know AWS specialists, have cash incentives, and develop their AWS abilities. You may save up to 72% compared to On-Demand pricing with Reserved Instances, and with Convertible Reserved Instances, you can change instance families, OS types, and tenancies. Reserved Instances also offer capacity reservations, so you can create reserved instances with certainty when you need to. Web-based Labs, USD 35 in AWS credits, and access to the AWS Educate Student Portal are all part of what students get with AWS Educate. Individuals of participating organizations can receive up to $100 in AWS credits through AWS Educate [28].

Microsoft Azure:

Organisations can achieve great cost savings in the short and long term across all of their resources—applications, storage, networking, data, servers, and data analytics—through Azure migration. By porting their current Windows Server, SQL Server, or Linux subscriptions to Azure, organisations may take advantage of Azure's hybrid benefits and optimise their hybrid environment costs. With the help of the Azure Migration and Modernization Programme, businesses can get free Azure credits to go towards migration expenses and to speed up their transition to the cloud with the help of consultants and other resources. Organisations can continue to receive extended security updates for three more years after the end of support dates when they transfer server workloads from 2008 and 2008 R2 to Azure.

Organizations receive direct assistance from Azure engineers for project development and deployment with expected incremental monthly usage of at least $5,000. Organizations can access discounted rates for development and testing on Azure with a Visual Studio subscription [29].

Organizations during the Microsoft Purview product preview, scan SQL Servers, and Power BI tenants at no cost. Organizations benefit from the Enterprise Agreement Support Plan Offer. They can save up to 65% on selected compute services with Azure's flexible and straightforward savings plan, tailored to their preferred plan term and hourly spending. All customers can explore Azure services with up to $150 in monthly credits through their Visual Studio subscription. At two- or four-year colleges, all full-time students who are 18 or older get $100 and several free services during their first year. Azure, Dynamics 365, and Microsoft 365 discounts, grants, and $3,500 per year are available to qualified customers and NGOs. Subscriptions to professional developer tools, software, and services are available to academic instructors and students at affordable prices. Businesses can save as much as 72 percent on certain Azure services, such virtual machines, when they pay in advance, with the option to pay monthly at no additional cost.

For interruptible workloads, benefit from up to 90% off pay-as-you-go pricing with Spot Virtual Machines, setting a maximum spending limit as needed [29].

Google Cloud Platform (GCP)

Google Cloud Platform (GCP) offers partner Advantage plans at three participation levels, each with distinct benefits tailored to help you sell, service, and innovate with Google Cloud. Upon joining Partner Advantage, organizations gain access to foundational benefits designed to enhance their skills and support their customers' success. Examples of benefits include discounted use of Google Workspace, technical training courses, participation in events, webinars, hands-on labs, and more. Upon achieving Partner status by completing specified tasks, they earn a Google Cloud Partner badge and unlock additional benefits. Examples of these benefits include partner discounts, financial incentives, inclusion in the Partner Directory, and more. Upon reaching the Premier Partner level, organizations earn a Google Cloud Premier Partner badge and gain access to the highest tier of benefits. Examples of Premier Partner benefits include enhanced discounts, increased financial incentives, exclusive webinars, advanced technical training, and more. All students can access free online labs, skill badges, and courses through Google Cloud Skills Boost, enabling them to follow a structured path or create their journey toward success in a cloud-centric environment. The Google Cloud Platform (GCP) student plan includes 200 free Google Cloud Skills Boost credits. These 200 Credits are valid for only one year from issuance [30].

VII. CONCLUSION

In this paper, we discussed the types of cloud services, cloud deployment models, leading cloud service providers, and key features of AWS, Azure, and GCP. When organizations choose between AWS, Azure, and GCP, they must consider their specific requirements, existing technology stack, expertise, and long-term strategy. Each cloud service provider has its strengths, and the best choice often depends on the unique needs of the business.

VIII. FUTURE WORK

However, selecting the right cloud service provider requires a thorough evaluation of multiple factors. Future efforts will focus more on expanding the comparison criteria for major cloud service providers like AWS, Azure, and GCP, as well as other providers in the market. This development intends to give extensive insights into the strengths and weaknesses of each cloud service provider, helping businesses make the right decisions based on their specific business requirements and objectives.

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